Module 2 - Robot Kinematics: Review and Assessment

Curriculum: Physical AI | Lessons: 6-10 | Recommended time: 20-30 minutes

Review checklist

- [ ] Distinguish links, revolute/prismatic joints, and DOF.
- [ ] Compute XY position with two-link FK.
- [ ] Test target reachability and distinguish the two IK elbow branches.
- [ ] Explain the Jacobian as a mapping from joint velocities to end-effector velocity.
- [ ] Explain why a singular Jacobian restricts Cartesian motion without freezing the joints.
- [ ] Describe the iterative numerical IK loop and its stopping criterion.

Part A - Knowledge check (1 point each)

Q1. A serial planar arm has three independent revolute joints. How many joint DOFs? (A) 1, (B) 3, (C) 6

Q2. Which best describes forward kinematics? (A) Joint coordinates -> end-effector pose, (B) Target pose -> joint coordinates, (C) Forces -> torques

Q3. For L1=3 m, L2=2 m, which target radial distance is geometrically unreachable? (A) 0.5 m, (B) 3 m, (C) 5 m

Q4. If a two-link planar Jacobian is singular, what follows? (A) All joints are locked, (B) Some instantaneous Cartesian velocity directions are unavailable, (C) Link lengths are zero

Q5. Why does numerical IK recalculate position error each iteration? (A) The current gripper position changes, (B) Link lengths change, (C) The Jacobian is always constant

Q6. Why can one reachable XY target have two analytical IK solutions? (A) Different elbow configurations, (B) Gravity changes length, (C) FK is random

Q7. In p_dot = J*q_dot, what are the units of p_dot for planar XY position? (A) rad/s, (B) m/s, (C) N*m

Q8. For L1=3, L2=2, theta1=0, theta2=90 deg, what is (x,y)? (A) (5,0), (B) (3,2), (C) (2,3)

Part B - Short calculations (2 points each)

Q9. FK: With L1=3, L2=2, theta1=0 deg, theta2=90 deg, calculate x and y, showing substitutions.

Q10. Reachability: For L1=4, L2=2, state the complete geometric radial reach interval. Is a target 1 m from the base reachable?

Q11. Jacobian velocity: Given J=[[-2,-2],[3,0]], q_dot=[0.1,0.2] rad/s, compute p_dot.

Q12. Singularity: Explain what det(J)=0 means for planar 2D instantaneous gripper motion, and why the joints may still move.

Part C - Practical verification (4 points)

Using your Lesson 10 numerical IK script, change the target to (3,2) with L1=3 and L2=2. Record: initial angles; final angles; iterations; final Cartesian error; whether the solver met its tolerance; and one screenshot of the trajectory. Joint angles can differ by elbow branch or equivalent angle wrapping. Briefly explain what changes if initialized near a singular configuration.

Scoring

• Part A: 8 points.

• Part B: 8 points.

• Part C: 4 points.

• Total: 20 points. Suggested strong-pass threshold: 16/20, with correct understanding of FK vs IK, reachability, and singularities.

Historical Module 2 assessment record

The completed in-chat assessment scored 3/5 initially. Incorrect concepts were FK vs IK and inner-bound reachability. A focused two-question reinforcement was completed with both answers correct, after which Module 2 was marked complete. This worksheet is an additional reusable review resource, not a claim that its questions have been attempted.

Instructor / Self-check Answer Key

Q1: B (3 independent joint coordinates).

Q2: A (FK maps joint coordinates to end-effector pose).

Q3: A (reachable 1 <= r <= 5; 0.5 m is too close).

Q4: B (Jacobian loses rank; not all instantaneous XY motions attainable).

Q5: A (the current state and therefore target error change).

Q6: A (positive/negative relative elbow angle for many targets).

Q7: B (m/s).

Q8: B ((3,2)).

Q9: x=3*cos(0)+2*cos(90 deg)=3, y=3*sin(0)+2*sin(90 deg)=2 meters.

Q10: abs(4-2) <= r <= 4+2, i.e. 2 <= r <= 6 meters; r=1 is unreachable.

Q11: p_dot=[-2*0.1-2*0.2, 3*0.1+0*0.2]=[-0.6,0.3] m/s.

Q12: Jacobian rank drops, so at least one Cartesian velocity direction is unavailable instantaneously; joints may still rotate and produce motion along remaining directions.

Part C rubric (4): 1 point successful target/error evidence; 1 correct program output; 1 trajectory screenshot; 1 accurate explanation of near-singular numerical sensitivity. Avoid promising convergence from every initial state.